

COLLEGE INSTITUTE OF SCIENCE & TECHNOLOGY,
BHOPAL

DEPARTMENT OF CSE

COURSE FILE

Programme : BE

Semester : VIII

Course Code : CS801

Subject Name : Soft Computing

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Approved By:

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SOFT COMPUTING CS801 SYLLABUS

Unit – I Soft Computing: Introduction of soft computing, soft computing vs. hard computing, various types of soft computing techniques, applications of soft computing. Artificial Intelligence : Introduction, Various types of production systems, characteristics of production systems, breadth first search, depth first search techniques, other Search Techniques like hill Climbing, Best first Search, A* algorithm, AO* Algorithms and various types of control strategies. Knowledge representation issues, Propositional and predicate logic, monotonic and non monotonic reasoning, forward Reasoning, backward reasoning, Weak & Strong Slot & filler structures, NLP.

Unit – II Neural Network : Structure and Function of a single neuron: Biological neuron, artificial neuron, definition of ANN, Taxonomy of neural net, Difference between ANN and human brain, characteristics and applications of ANN, single layer network, Perceptron training algorithm, Linear separability, Widrow & Hebb's learning rule/Delta rule, ADALINE, MADALINE, AI v/s ANN. Introduction of MLP, different activation functions, Error back propagation algorithm, derivation of BBPA, momentum, limitation, characteristics and application of EBPA,

Unit – III Counter propagation network, architecture, functioning & characteristics of counter Propagation network, Hopfield/ Recurrent network, configuration, stability constraints, associative memory, and characteristics, limitations and applications. Hopfield v/s Boltzman machine. Adaptive Resonance Theory: Architecture, classifications, Implementation and training. Associative Memory.

Unit – IV Fuzzy Logic: Fuzzy set theory, Fuzzy set versus crisp set, Crisp relation & fuzzy relations, Fuzzy systems: crisp logic, fuzzy logic, introduction & features of membership functions, Fuzzy rule base system : fuzzy propositions, formation, decomposition & aggregation of fuzzy rules, fuzzy reasoning, fuzzy inference systems, fuzzy decision making & Applications of fuzzy logic.

Unit – V Genetic algorithm : Fundamentals, basic concepts, working principle, encoding, fitness function, reproduction, Genetic modeling: Inheritance operator, cross over, inversion & deletion, mutation operator, Bitwise operator, Generational Cycle, Convergence of GA, Applications & advances in GA, Differences & similarities between GA & other traditional methods.

References :

S, Rajasekaran & G.A. Vijayalakshmi Pai, Neural Networks, Fuzzy Logic & Genetic Algorithms, Synthesis & applications, PHI Publication.

S.N. Sivanandam & S.N. Deepa, Principles of SoftComputing, Wiley Publications
Rich E and Knight K, Artificial Intelligence, TMH, New Delhi.

Bose, Neural Network fundamental with Graph , Algo.& Appl, TMH

Kosko: Neural Network & Fuzzy System, PHI Publication

Klir & Yuan ,Fuzzy sets & Fuzzy Logic: Theory & Appli.,PHI Pub.

(ii) Lecture Plan with References

Subject Title : Soft Computing	Session : 2011:2012
Subject Code : CS801	Semester : VIII
Department : Computer Science & Engg	Branch : Computer Science & Engg.

L. No.	Topics to be covered	Unit	Completion Date	Reference
1.	Soft Computing : Introduction of soft computing	I		R2:8
2.	soft computing vs. hard computing,	I		R2:8, R2:8
3.	various types of soft computing techniques Applications of soft computing.	I		R2:3
4.	Artificial Intelligence : Introduction, Various types of production systems	I		R1:1, R1:30
5.	characteristics of production systems, breadth first search	I		R1:43, R1:32
6.	depth first search techniques, other Search Techniques like hill Climbing	I		R1:32, R1:52
7.	Best first Search, A* algorithm	I		R1:57, R1:59
8.	AO* Algorithms and various types of control Strategies.	I		R1:67
9.	Knowledge representation issues, Propositional and predicate logic	I		R1:79, R1:98
10.	monotonic and non monotonic reasoning, forward Reasoning, backward reasoning	I		R1:147, R1:134
11.	Weak & Strong Slot & filler structures, NLP	I		R1:188, R1:285
12.	Neural Network : Structure and Function of a single neuron	II		R2:11
13.	Biological neuron, artificial neuron	II		R2:12
14.	definition of ANN, Taxonomy of neural net	II		R2:24
15.	Difference between ANN and human brain, characteristics and applications of ANN	II		R2:14, R2:3
16.	single layer network, Perceptron training algorithm	II		R2:49
17.	Linear separability, Widrow & Hebb;s learning rule/Delta rule	II		R2:29, R2:31
18.	ADALINE, MADALINE	II		R2:57, R2:60
19.	AI v/s ANN, Introduction of MLP	II		R2:64
20.	different activation functions, Error back propagation algorithm	II		R2:64
21.	derivation of BBPA, momentum, limitation, characteristics and application of EBPA	II		R2:70
22.	Counter propagation network, architecture	III		R2:165
23.	functioning & characteristics of counter Propagation network	III		R2:173

24.	Hopfield/ Recurrent network, configuration, stability constraints	III		R2:110
25.	Associative memory, and characteristics, limitations and applications	III		R2:97
26.	Hopfield v/s Boltzman machine.	III		R2:97, R2:233
27.	Adaptive Resonance Theory: Architecture, classifications,	III		R2:178
28.	Implementation and training. Associative Memory.	III		R2:181
29.	Fuzzy Logic: Fuzzy set theory	IV		R2:251
30.	Fuzzy set versus crisp set, Crisp relation & fuzzy relations	IV		R2:255, R2:273
31.	Fuzzy systems: crisp logic, fuzzy logic	IV		R2:255, R2:273
32.	introduction & features of membership functions	IV		R2:295
33.	Fuzzy rule base system : fuzzy propositions	IV		R2:347
34.	formation, decomposition & aggregation of fuzzy rules	IV		R2:352
35.	fuzzy reasoning, fuzzy inference systems	IV		R2:353, R2:355
36.	fuzzy decision making & Applications of fuzzy logic	IV		R2:363
37.	Genetic algorithm : Fundamentals	V		R2:385, R3:228
38.	basic concepts, working principle	V		R2:386, R3:230
39.	encoding, fitness function, reproduction	V		R2:405, R2:399
40.	Genetic modeling: Inheritance operator	V		R3:253
41.	cross over, inversion & deletion, mutation operator	V		R3:254
42.	Bitwise operator, Generational Cycle	V		R3:265
43.	Convergence of GA, Applications & advances in GA	V		R3:271
44.	Differences & similarities between GA & other traditional methods	V		R3:287

Reference Books:

R1: Artificial intelligence by Rich and Knight, Third Edition, TMH
R2: Principles of Soft Computing by Sivanandam & Deepa, Second Edition, Wiley india
R3: Neural Networks, Fuzzy Logic and Genetic Algorithms: Synthesis and Applications
by S. Rajasekaran, G. A. Vijayalakshmi Pai, Wiley india

TUTORIAL SHEET 1

1. What do you mean by artificial neural networks? An artificial neural network is an information processing system that has certain performance characteristics in common with biological neural networks.
2. What is neural network architecture? A neural network is characterized by its pattern of connections between the neurons called its architecture.
3. What is meant by training of artificial neural networks?
The method of determining the weights on the connections called training.
4. What is meant by weights?
The weights represent information being used by the net to solve problem.
5. Write the logistic sigmoid function?
$$f(x) = 1/(1+\exp(-x)).$$
6. What are the important characteristics that artificial neural networks share with biological neural systems? Fault tolerance.
7. Name some applications of artificial neural networks. Signal processing, Control, pattern recognition, Medicine, Speech production, Speech recognition and Business etc...
8. What is the classification of training? Supervised and unsupervised.
9. What is supervised training? Training is accomplished by presenting a sequence of training vectors or patterns, each with an associated target output vector. The weights are adjusted according to the learning algorithm. This process is known as supervised training.
10. What is associative memory? A neural net that is trained to associate a set of input vectors with a corresponding set of output vectors is called associative memory.
11. What is auto associative memory? If the desired output vector is same as the input vector, the net is an auto associative memory.
12. What is hetero-associative memory? If the desired output vector is different from the input vector, the net is a hetero-associative memory.
13. What is unsupervised training? A sequence of input vectors is provided, but no target vectors are specified. The net modifies the weights so that the most similar input vectors are assigned to the same output unit.
14. Define identity function.
$$f(x) = x \text{ for all } x.$$
15. Define binary step function with threshold .
$$f(x) = 1 \text{ if } x \geq \theta$$
$$f(x) = 0 \text{ if } x < \theta$$
16. What are the three layers of perceptron? Sensory units, associator units, and a response unit.
17. Define delta rule. The delta rule changes the weights of the neural connections so as to minimize the difference between the net input to the output unit and the target value.
18. Draw the structure of ADALINE net.
19. Draw the structure of MADALINE net.
20. Draw the structure of back propagation net.

TUTORIAL SHEET 2

21. What is unsupervised training? A sequence of input vector is provided, but no target vectors are specified. The net modifies the weights so that the most similar input vectors are assigned to the same output unit.
22. Which net is related to “winner take all”? Competitive learning network.
23. What is meant by winner take all? Only the neuron with the largest activation is allowed to remain on.
24. Draw the architecture of MAXNET.

25. What is the equation for updating the activation?

$$A_{j \text{ (new)}} = f[a_{j \text{ (old)}} - a_{k \text{ (old)}}]$$

26. Draw the architecture of Mexican cat.

Draw the architecture.

27. What is the activation unit x_i at time t of Mexican cat?

$$X_i(t) = f[S_i(t) + w_{i+k} x_{i+k}(t-1)]$$

28. What is hamming net?

A hamming net is a maximum likelihood classifier net that can be used to determine which of the exemplar vector is most similar to an input vector.

29. Define hamming distance between two vectors.

It is the number of components in which vector differ.

30. Draw the architecture of hamming bet. Draw the architecture.

31. What is the other name of Kohonen Self-organising maps? Topology preserving maps.

32. What is the concept of hebbian learning? Hebb proposed that learning occurs by modification of the synapse strengths(weights) in a manner such that if two interconnected neurons are both on at the same time, then the weights between these neurons should be increased.

33. Draw the architecture of Adaptive Resonance Theory.

34. What is the other name of Auto associative net? Content addressable nature.

35. What is stability-plasticity dilemma? A learning agent should be plastic, or adaptive in reacting to changing environments meanwhile it should be stable to preserve knowledge acquired previously.

36. What is leaky learning? To update the weights of both the winning and losing units, but use a significantly smaller learning rate for the losers; this is commonly referred to as leaky learning.

37. Name some application of competitive learning network. Data compression in communication and image processing, graph partitioning and word perception models.

38. Name some application of Kohonen self-organizing network . neural phonetic type writer, to learn ballistic arm movements.

39. What is general content addressable memory? Any physical system whose dynamics in phase space is dominated by a substantial number of locally stable states to which it is attracted can therefore be regarded as general content-addressable memory.

40. What is basin of attraction? The present input pattern vector cannot escape from a region.

TUTORIAL SHEET 3

41. In which situation fuzzy logic is most suitable.

i) Very complex models where understanding is strictly limited or in fact quite judgmental.

ii) process where human reasoning, human perception, or human decision making are inextricably involved.

42. What is called the principle of incompatibility?

Conventional techniques for system analysis are intrinsically unsuited for dealing with humanistic systems, whose behavior is strongly influenced by human judgment, perception and emotion. This is the manifestation of what might be called the principle of incompatibility.

43. Write an example for linguistic variable and values. In the sentence “age is young”, age is the linguistic variable and young is the linguistic value.

44. Define the concentration of linguistic values. Let A be the linguistic value, then the operation of concentration is defined as $CON(A) = A^2$

45. Define the dilation of linguistic values. Let A be the linguistic value, then the operation of dilation is defined as $DIL(A) = 0.5 A$

46. Define contrast intensification.

$$INT(A) = \begin{cases} A^2, & \text{for } 0 \leq \mu_A(x) \leq 0.5 \\ 2(A), & \text{for } 0.5 \leq \mu_A(x) \leq 1 \end{cases}$$

47. Write the Material implication of A entails B

$$R = A \rightarrow B = A \cup B$$

48. Write the Proportional calculus of A entails B

$$R = A \rightarrow B = A \cup (A \rightarrow B)$$

49. Write the Extended Proportional calculus of A entails B

$$R = A \rightarrow B = (A \rightarrow B) \cup B$$

50. Write the Generalization of modus ponens of A entails B

$$\mu_R(x,y) = \sup \{c \mid \mu_A(x) * c \leq \mu_B(y) \text{ and } 0 \leq c \leq 1\}.$$

51. What is classical set? A set with crisp boundary.

52. What is fuzzy set? A set without crisp boundary.

53. What is membership function? If X is a collection of objects denoted generically by x, then a fuzzy set A in X is defined as a set of ordered pairs $A = \{(x, \mu_A(x)) \mid x \in X\}$

where $\mu_A(x)$ is called the membership function for the fuzzy set A.

54. Define Support. The support of the fuzzy set A is the set of all points x in X such that

$$\mu_A(x) > 0. \text{ Support}(A) = \{x \mid \mu_A(x) > 0\}$$

55. Define Core. A core of a fuzzy set A is the set of all points x in X such that $\mu_A(x) = 1$.

$$\text{Core}(A) = \{x \mid \mu_A(x) = 1\}$$

56. Define normality. A fuzzy set A is normal if its core is non empty. In other words, we can always find a point $x \in X$ such that $\mu_A(x) = 1$.

57. Define Cross over point. A cross over point of a fuzzy set A is a point $x \in X$ at which

$$\mu_A(x) = 0.5 \text{ Crossover}(A) = \{x \mid \mu_A(x) = 0.5\}$$

58. Define fuzzy singleton. A fuzzy set whose support is a single point in X with $\mu_A(x) = 1$ is called a fuzzy singleton.

TUTORIAL SHEET 4

60. What is the name of parameters in the layer 1 of ANFIS?
Premise parameter
61. What is the output of layer3 of ANFIS?
Normilised firing strength.
62. What is the name of parameters in the layer4 of ANFIS?
Consequent parameters.
63. What is the acronym for CANFIS?
Coactive neuro-fuzzy inference systems.
64. What are the applications of K-means clustering? Image and speech data compression, data preprocessing for system modeling using radial basis function networks and task decomposition in heterogeneous neural network architectures.
65. What are the applications of fuzzy C-means clustering? Medical image segmentation and qualitative modeling.
66. What is mountain clustering method? Approximate estimation of cluster centers on the basis of a density measure called mountain function.
67. What is subtractive clustering?
Here the data points are considered as the candidates for cluster center.
68. What is balanced –sampling criterion? Cut the dimension in which the training data associated with the region are most spread out and cut it as the median value of those samples in that dimension.
69. What is direct evaluation?
To feed the structure into the parameter identification phase and use the final performance to choose the best cut.
70. Define a binary fuzzy boxtree.
It is rooted tree in which each internal node has two children.
71. What is knowledge acquisition?
Acquisition of human operator's knowledge about how to control a system and generates a set of fuzzy if-then rules as the backbone for a fuzzy controller that behaves like the original human operator.
72. What is linguistic information?
An experienced human operator can usually summarize his or her reasoning process in arriving at final control actions or decisions as a set of fuzzy if-then rules with imprecise but correct membership functions.
73. What is numerical information?
When a human operator is working, it is possible to record the sensor data observed by the human and the human's corresponding actions as a set of desired input output data pairs.
74. What is stage adaptive network?
The adaptive network containing the FC block and the plant block are referred as stage adaptive network.
75. What is fuzzy inference systems?
It is a popular computing frame work based on the concepts of fuzzy set theory, fuzzy if-then rule, and fuzzy reasoning.
76. What are the three conceptual components of the basic structure of a fuzzy inference system? Rule base, data base and reasoning mechanism.

TUTORIAL SHEET 5

77. What is genetic algorithm? Genetic algorithms are search algorithms based on the mechanics of natural selection and natural genetics.
78. What is the theme of research on genetic algorithms? The central theme of research on genetic algorithms has been robustness, the balance between efficiency and efficacy necessary for survival in many different environments.
79. Name some of the existing search methods. Calculus based methods, enumerative schemes, random search algorithms.
80. What are the operators involved in a simple genetic algorithm? Reproduction, cross over, mutation.
81. What is reproduction? Reproduction is a process in which individual strings are copied according to their objective function.
82. What is the use of SCHEMATA? The schemata provides a tools to answer the questions such as how one string can be similar to its fellow strings and in what ways is a string a representative of other string classes with similarities at certain string positions.
83. What is schema?
A schema is a similarity template describing a subset of strings with similarities at certain string positions.
84. What is cross over?
Cross over is a recombination operator.
85. What are the types of cross over? Single site cross over, two point cross over, multipoint cross over and uniform cross over.
86. What are the types of multi point cross over?
Even number of cross sites and odd number of cross sites.
87. What is cross over rate?
Cross over rate is usually denoted as P_c the probability of cross over.
88. What is inversion?
The string from the population is selected and the bits between two random sites are inverted.
89. What are the types of inversion?
Linear+end-inversion , continuous inversion and mass inversion.
90. What is deletion and duplication? Any two or three bits at random in order are selected and the previous bits are duplicated.
91. What is segregation?
The bits of the parents are segregated and then crossed over to produce offspring.
92. What is cross over and inversion?
It is the combination of both cross over and inversion.

UNIT TEST 1

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UNIT TEST 2

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UNIT TEST 3

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MID SEM PAPER

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